Enterprise Infrastructure Plan

ABC Startup Solutions, Inc.

Course: ITEP 414 - System Administration and Maintenance

Program: Bachelor of Science in Information Technology

Week: 2 - Portfolio Activity

Prepared by: Joya, Mazel Catherine P. (Junior System Administrator)

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PART 1 - Company Profile

Company Name

ABC Startup Solutions, Inc.

Nature of Business

ABC Startup Solutions is a newly established software development company that designs and delivers custom web applications, mobile applications, and cloud-based business systems for small and medium-sized enterprises. The company focuses on helping businesses transition into the digital space by providing tailored software that fits their operations, from inventory management systems to customer-facing mobile apps. As a startup, they are still in their early growth stage, which means every IT decision will directly impact how quickly and effectively they can scale.

Company Vision

To be the most trusted digital transformation partner of Filipino businesses by delivering innovative, reliable, and scalable software solutions. The company envisions a future where even the smallest sari-sari store or local shop can benefit from custom-built technology, and they want to be the ones making that happen.

Company Mission

To empower businesses through intelligent software solutions, foster a culture of innovation and collaboration among our employees, and deliver value to our clients with integrity, quality, and excellence. The company believes that great software starts with a great team, which is why they prioritize open communication, continuous learning, and a supportive work environment for all 20 members of the organization.

Office Location (Fictional)

5th Floor, CyberOne Tower, Ayala Avenue, Makati City, Metro Manila, Philippines. The office occupies a single open-plan floor with designated areas for each department, a server room at the back, and a small meeting room near the entrance. The location was chosen for its proximity to business clients and accessible public transportation.

Organizational Structure

The company follows a relatively flat organizational structure typical of startups, which allows for faster decision-making and more direct communication between departments. The hierarchy looks like this:

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CEO / President
|-- Chief Technology Officer (CTO)
|   |-- IT Department Head
|   |   |-- System Administrator (Junior) <-- You are here
|-- HR Manager
|-- Finance Manager
|-- Sales Manager
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As the Junior System Administrator, I report directly to the IT Department Head, who in turn reports to the CTO. This means my work directly supports the technology backbone of the entire company, and any issue I resolve or system I maintain has a ripple effect on every employee.

Employee Distribution

Department	Number of Employees	Primary Function
Information Technology	5	Software development, infrastructure management, technical support
Human Resources	4	Recruitment, payroll processing, employee relations, compliance
Finance	5	Accounting, budgeting, billing, financial reporting
Sales	6	Client acquisition, marketing proposals, account management
TOTAL	**20**	

The IT department has the fewest dedicated employees but supports the entire organization's technical needs. The Sales team is the largest because client-facing roles require more manpower to handle multiple accounts simultaneously.

PART 2 - Enterprise Hardware Inventory

All hardware was selected to support 20 employees performing software development, administrative, and sales functions. The selection process involved considering each department's specific workload, the physical layout of the office, and the company's budget constraints as a startup. Desktops are prioritized for IT, HR, and Finance employees who need fixed, high-performance workstations at their desks, while laptops are assigned to the Sales team who frequently visit clients and need portability for presentations. Quantities include spare units and extra monitors to ensure business continuity and prepare for minor expansion.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer (Intel i5-13400, 16GB RAM, 512GB NVMe SSD)	15	IT (4), HR (3), Finance (4), Sales (4)	Daily workstation for software development, documentation, data entry, and general office tasks
HW-002	Laptop Computer (Intel i5-1335U, 16GB RAM, 512GB NVMe SSD)	5	Sales	Portable workstations for client presentations, on-site demos, and fieldwork
HW-003	Dell PowerEdge T340 Tower Server (Xeon E-2334, 32GB ECC RAM, 2x1TB NVMe SSD, 2x4TB HDD)	1	IT (Server Room)	Central server hosting file server, application server, and PostgreSQL database

Asset ID	Hardware	Quantity	Department	Purpose
HW-004	Cisco RV345 Dual WAN Gigabit VPN Router	1	IT (Server Room)	Connects office LAN to ISP, routes traffic, supports VPN for remote access
HW-005	Cisco Catalyst 2960-X 48-Port Managed Switch	2	IT (Server Room)	Distributes network connectivity to all workstations, server, NAS, and printers
HW-006	HP LaserJet Pro M404dn Network Printer	2	HR (1), Finance (1)	Printing contracts, reports, payslips, and official documents
HW-007	APC Back-UPS Pro 1500VA / 900W UPS	21	All Departments	Battery backup and surge protection for 20 workstations and 1 server
HW-008	Cisco Aironet 1850i Wireless Access Point	3	Office Floor (distributed)	Wi-Fi coverage for laptops, mobile phones, and guest devices
HW-009	Synology DiskStation DS920+ NAS (4-Bay, 2x4TB Seagate IronWolf)	1	IT (Server Room)	Centralized file storage, sharing, and backup target
HW-010	Seagate Backup Plus Hub 4TB External Hard Drive	2	IT (Server Room)	Offline backup copies following the 3-2-1 backup rule
HW-011	Dell P2422H 24" Full HD LED Monitor	25	All Departments	Primary displays; IT developers get dual-monitor setup

Detailed Justification: Starting a company from zero means every single piece of hardware must be carefully chosen. I selected 15 desktop units because those employees work primarily at fixed desks, while the 5 Sales employees need laptops for client visits. The single Dell PowerEdge T340 server is cost-effective for a 20-person startup. It is powerful enough to host file services, run a database, and even virtualize the pfSense firewall, without the overhead of maintaining multiple physical servers. Two Cisco 2960-X switches with 48 ports each provide 96 total ports, more than enough for 20 workstations plus servers, printers, and access points, while leaving room for future growth. Twenty-one UPS units are non-negotiable because the Philippines experiences frequent power fluctuations, and unprotected equipment can suffer permanent damage. Two external backup drives implement the 3-2-1 rule: three copies of data, on two different media (NAS and external drive), with one copy kept offsite.

PART 3 - Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	24H2	Commercial (Volume License)	Standard desktop OS for all 20 workstations; Pro edition supports BitLocker encryption, domain join, and Group Policy
Ubuntu Server	22.04 LTS	Open Source (Free, GPL)	Server OS for Dell PowerEdge T340; chosen for stability, 5-year LTS support, and zero licensing cost
Microsoft Office	Microsoft 365 Business Standard	Subscription (per user/month)	Word, Excel, PowerPoint, Outlook, and Teams for document creation and collaboration
Visual Studio Code	1.9x (latest stable)	Free / Open Source (MIT)	Primary code editor for software development with Git and GitHub integration
Git	2.4x (latest stable)	Free / Open Source (GPL)	Distributed version control for tracking source code changes
GitHub Desktop	3.x (latest stable)	Free (Freeware)	GUI client for Git; simplifies branching, committing, and pushing to GitHub

Software	Version	License	Purpose
VirtualBox	7.x (latest stable)	Free / Open Source (GPL v3)	Virtualization for testing environments and running the pfSense firewall VM
Google Chrome	Latest stable	Freeware	Standard web browser for all employees; simplifies IT support
Microsoft Defender	Latest (built-in)	Included with Windows 11 Pro	Real-time antivirus and anti-malware at zero additional cost
AnyDesk	Latest stable	Free (Personal) / Commercial	Remote desktop support for troubleshooting employee workstations
7-Zip	24.x (latest stable)	Free / Open Source (LGPL)	File compression and archive management

Why each software is needed:

- **Windows 11 Pro** is the backbone of every workstation. The Pro edition is required because Home lacks BitLocker encryption and domain join capabilities, both essential for a managed business network.
- **Ubuntu Server** saves thousands of pesos in licensing fees while providing the same (or better) stability and security for hosting file services, databases, and virtual machines.
- **Microsoft Office 365** handles all business documents, communications, and scheduling. The cloud subscription model provides automatic updates and no manual patching.
- **Visual Studio Code** is the industry standard for modern software development, with its extension ecosystem, integrated terminal, and GitHub integration.
- **Git and GitHub Desktop** enforce professional version control from day one, preventing code loss and enabling team collaboration.
- **VirtualBox** gives the IT team a sandbox for testing new software in isolated virtual machines before deploying to production.
- **Google Chrome** ensures consistency across all workstations, simplifying IT support and security updates.
- **Microsoft Defender** provides free baseline antivirus protection with cloud-powered threat detection.
- **AnyDesk** is essential for the lone Junior System Administrator, allowing remote troubleshooting of 20 workstations without walking to each desk.
- **7-Zip** handles compressed files from software development, backups, and file sharing.

PART 4 - Enterprise Network Inventory

Item	Model / Specification	Quantity	Purpose
ISP Modem (ONT)	PLDT Fibr GPON ONT (provided by ISP)	1	Converts fiber optic signal into Ethernet connection for the router
Router	Cisco RV345 Dual WAN Gigabit VPN Router	1	Routes traffic between office LAN and internet; supports backup ISP and VPN
Firewall	pfSense (virtual appliance on Ubuntu Server)	1	Inspects, filters, and controls all inbound/outbound traffic

Item	Model / Specification	Quantity	Purpose
Managed Switch	Cisco Catalyst 2960-X 48-Port Gigabit	2	Wired connectivity for all devices; supports VLAN segmentation
Wireless Access Point	Cisco Aironet 1850i (802.11ac Wave 2)	3	Wireless coverage across the office floor
Patch Panel	Cat6 24-Port Rackmount	2	Centralized termination for structured cabling
CAT6 Ethernet Cable	UTP, 350MHz, 305m Pull Box	2 boxes (610m total)	Structured cabling from patch panels to workstations
RJ45 Connectors	Cat6 RJ45 Pass-Through (bag of 100)	500	Terminating cable runs at patch panels and outlets

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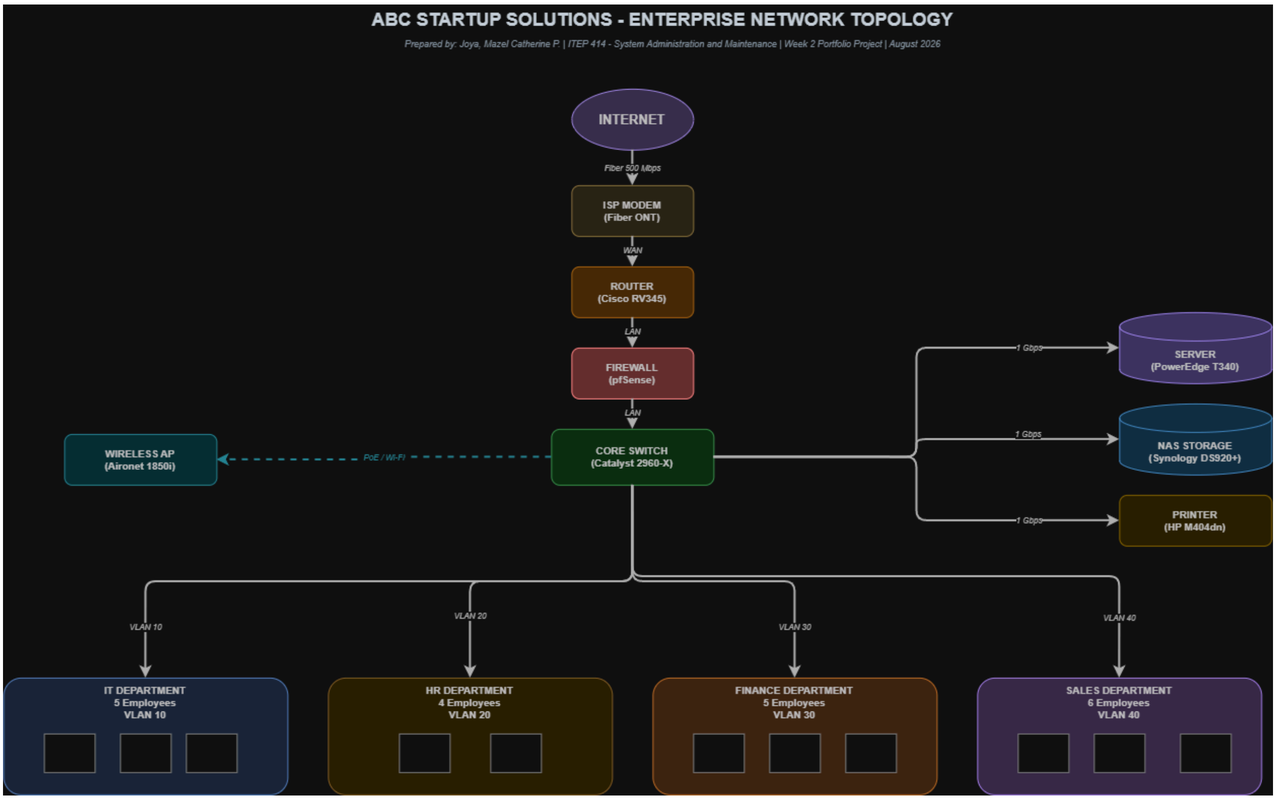
PART 5 - Enterprise Network Diagram

The network follows a star topology centered on the core switches. All internet traffic flows from the ISP modem through the router and firewall before reaching the internal network, ensuring every packet is inspected. The server, NAS, and printers connect directly to the switch, while three access points provide wireless coverage for laptops and mobile devices. Departments are logically separated using VLANs to improve security and performance. For example, the Sales team's traffic cannot reach the Finance department's network segment unless explicitly permitted by firewall rules.



- **Diagram Title:** ABC Startup Solutions, Enterprise Network Topology
- **Designed by:** Joya, Mazel Catherine P. (BSIT Student)
- **Tool:** Draw.io (editable version in `diagrams/network-topology.drawio`)
- **Exports:** PNG and PDF available in `diagrams/` folder

PART 6 - System Administration Roles

Understanding the different roles within a System Administration team is crucial because no single person can handle everything in a modern IT environment. Each role has its own specialized responsibilities, but they all share the same ultimate goal: keeping the organization's technology running smoothly and securely. Below I explain the four core IT roles relevant to a startup like ABC Startup Solutions.

Helpdesk Technician

The Helpdesk Technician is the first point of contact when any employee in the company encounters a technical problem. Whether someone's printer stops working, their email won't sync, or they accidentally locked themselves out of their account, the Helpdesk Technician is the person they call. This role requires strong customer service skills because the technician must communicate with non-technical employees patiently and clearly, often explaining solutions in simple terms.

Responsibilities: The Helpdesk Technician's daily tasks include answering support calls and emails, creating and managing support tickets in a ticketing system (such as Jira Service Management or Zendesk), troubleshooting basic hardware issues like monitor connections and keyboard malfunctions, resolving software problems like application crashes and update failures, resetting passwords, and escalating complex issues that require deeper technical expertise to the Network Administrator or System Administrator.

Skills: Success in this role demands excellent communication skills (both written and verbal), patience when dealing with frustrated users, basic Windows and macOS troubleshooting knowledge, familiarity with common office applications, problem-solving ability, and good time management to handle multiple tickets simultaneously.

Common Tools: Help desk ticketing systems (Jira Service Management, Zendesk, Freshdesk), remote support tools (AnyDesk, TeamViewer), Windows built-in diagnostic utilities (Event Viewer, Task Manager, Device Manager), and basic networking commands (ipconfig, ping, nslookup).

Certifications: CompTIA A+ (the industry standard for entry-level IT support), CompTIA IT Fundamentals (ITF+) for foundational knowledge, and Microsoft 365 Fundamentals (MS-900) for understanding cloud-based business tools.

Network Administrator

The Network Administrator is responsible for the company's entire network infrastructure, from the routers and switches in the server room to the Wi-Fi access points on the office floor. If the internet goes down, if employees cannot access shared files, or if a new department needs its own network segment, the Network Administrator is the person who makes it happen. This role is more technical than the Helpdesk position and requires deep knowledge of networking protocols and hardware configuration.

Responsibilities: The Network Administrator designs, configures, and maintains the LAN/WAN infrastructure, manages routers, switches, firewalls, and VLANs, monitors network performance and uptime using tools like Nagios or SolarWinds, implements access control lists and security policies, troubleshoots connectivity issues across the entire network, and plans for future network expansion as the company grows.

Skills: This role requires strong knowledge of TCP/IP, DNS, DHCP, subnetting, routing and switching protocols, firewall rule configuration, network monitoring and analysis, structured cabling standards, and the ability to read and create network diagrams.

Common Tools: Cisco IOS (for configuring switches and routers), pfSense (for firewall management), Wireshark (for packet analysis and troubleshooting), PuTTY or Termius (for SSH connections to network devices), and monitoring platforms like SolarWinds or Nagios.

Certifications: CompTIA Network+ (foundational networking knowledge), Cisco Certified Network Associate (CCNA) (the gold standard for network professionals), and Juniper Networks Certified Associate (JNCIA) for multi-vendor environments.

Linux System Administrator

The Linux System Administrator is the person who keeps the company's servers running. In a startup like ABC Startup Solutions, the Linux SysAdmin manages the Dell PowerEdge server running Ubuntu, ensuring that file services, databases, and applications are always available. This role is deeply technical and requires comfort with the command line, because most server management tasks are performed through a terminal rather than a graphical interface.

Responsibilities: The Linux System Administrator installs and configures Ubuntu Server, manages user accounts and file permissions, writes shell scripts to automate repetitive tasks (like backups or log rotation), applies security patches and system updates, maintains web servers (Apache or Nginx), database servers (PostgreSQL or MySQL), and file sharing services (Samba or NFS), and monitors server performance to prevent bottlenecks.

Skills: This role demands proficiency in the Linux command line, Bash shell scripting, package management (apt), service management (systemd), file permissions and ownership, basic networking (IP configuration, firewall rules with iptables or ufw), and familiarity with common server applications.

Common Tools: SSH (for remote server access), vim or nano (for text editing), rsync (for file synchronization), cron (for scheduling automated tasks), systemctl (for managing services), Apache/Nginx (for web hosting), and MySQL/PostgreSQL (for database management).

Certifications: CompTIA Linux+ (foundational Linux knowledge), Red Hat Certified System Administrator (RHCSA) (the most respected Linux certification), and Linux Professional Institute Certification (LPIC-1) for vendor-neutral Linux expertise.

Cloud Administrator

The Cloud Administrator manages the company's cloud resources, including virtual machines, storage buckets, databases, and identity access management in platforms like AWS or Azure. As startups increasingly move their workloads to the cloud, this role becomes essential for scaling infrastructure without purchasing physical hardware. Even if the company currently uses an on-premises server, the Cloud Administrator prepares the path for eventual cloud migration.

Responsibilities: The Cloud Administrator provisions and manages cloud resources (virtual machines, storage, networking), configures identity and access management (IAM) to control who can access what, monitors cloud costs and usage to prevent budget overruns, ensures high availability and disaster recovery in the cloud, and manages cloud-native services like load balancers, auto-scaling groups, and managed databases.

Skills: This role requires understanding of cloud service models (IaaS, PaaS, SaaS), virtualization concepts, infrastructure-as-code (Terraform, CloudFormation), cost management and optimization, scripting (Python, Bash), and cloud security best practices (encryption, IAM policies, network security groups).

Common Tools: AWS Management Console / Azure Portal (for GUI-based management), AWS CLI / Azure CLI (for command-line operations), Terraform (for infrastructure-as-code), CloudWatch / Azure Monitor (for monitoring and alerting), and Git (for version-controlling infrastructure code).

Certifications: AWS Certified SysOps Administrator (for Amazon cloud expertise), Microsoft Azure Administrator (AZ-104) (for Microsoft cloud expertise), and Google Cloud Associate Cloud Engineer (for GCP expertise).

How These Professionals Work Together

Inside one organization, the Helpdesk Technician, Network Administrator, Linux System Administrator, and Cloud Administrator form a layered IT team that keeps the business running. Think of it like a hospital: the Helpdesk Technician is the triage nurse who sees every patient first, the Network Administrator is the cardiologist who keeps the circulatory system flowing, the Linux System Administrator is the surgeon who operates on the core systems, and the Cloud Administrator is the specialist who handles advanced procedures.

In practice, the Helpdesk Technician acts as the front line, resolving common issues directly, such as resetting a password, fixing a printer connection, or walking someone through an application error. When a problem is beyond their expertise, such as a network outage affecting the entire floor or a server crash, they escalate to the appropriate specialist. The Network Administrator ensures that every device, from the firewall to the Wi-Fi access points, can communicate securely and efficiently. When the Cloud Administrator spins up a virtual server in AWS, it must integrate cleanly with the company's network and VPN, which is only possible if the Network Administrator has configured the routing and firewall rules correctly.

The Linux System Administrator maintains the on-premises Ubuntu server, ensuring that file services, databases, and applications are always available. They often work closely with the Cloud Administrator to migrate or synchronize workloads between the local data center and the cloud, creating a hybrid environment that balances cost and scalability. Communication flows continuously among the four: the Helpdesk Technician reports recurring patterns it observes in user tickets (for example, "three employees this week reported slow file access"), the Network Administrator shares traffic data that reveals performance bottlenecks, and the two server administrators coordinate patches and backups so the whole team ships updates without downtime.

What makes this collaboration work is mutual respect and clear communication. The Helpdesk Technician understands that escalating a ticket is not a sign of weakness but a necessary step. The Network Administrator knows that the Helpdesk's ticket data is valuable intelligence for identifying infrastructure problems. The Linux and Cloud administrators understand that their work directly affects every employee's productivity, so they schedule maintenance windows during off-hours and communicate changes in advance. Their combined effort ensures a single, reliable IT service where incidents are handled quickly, infrastructure scales with the business, and no specialist works in isolation.

PART 7 - Infrastructure Recommendations

Internet Provider

Recommendation: PLDT Fibr Enterprise Plan, 500 Mbps symmetric fiber with a static IP address.

Reason: A software development company like ABC Startup Solutions continuously uploads code repositories, pushes large files to GitHub, runs video calls with clients, and may eventually host test environments that require incoming connections. All of these activities demand high upload speeds, which is why a symmetric fiber plan (equal upload and download) is essential. PLDT Fibr provides fiber-optic stability with minimal latency, and the static IP address is required for secure VPN access, remote desktop support, and hosting test servers that clients can access. A business-grade plan also includes a service-level agreement (SLA) that guarantees faster repair response times compared to residential plans. If the internet goes down at 2 PM on a workday, the company cannot afford to wait until the next day for a technician to arrive.

Server Specifications

Recommendation: Dell PowerEdge T340 Tower Server with Intel Xeon E-2334 (4-core, 3.4 GHz base), 32GB ECC DDR4 RAM, 2x 1TB NVMe SSD in RAID 1 (operating system and applications), and 2x 4TB HDD in RAID 1 (data storage and backups).

Reason: A single tower server is the most cost-effective choice for a 20-employee startup, but it must be reliable because it hosts the company's file server, application server, and PostgreSQL database, essentially the entire digital backbone of the business. ECC (Error Correcting Code) RAM is critical because it detects and corrects memory errors in real-time, preventing silent data corruption that could go unnoticed for weeks. RAID 1 mirroring on both the SSDs and HDDs ensures that if any single drive fails, the data remains accessible on its mirror pair with zero downtime. The 32GB of RAM provides headroom for running virtual machines (the pfSense firewall and test environments) alongside the primary services, so the server does not become a bottleneck during peak usage.

Backup Strategy

Recommendation: Implement the 3-2-1 backup rule rigorously: maintain 3 copies of all critical data, stored on 2 different types of media, with 1 copy kept offsite. Specifically:

- **Daily incremental backups** at 11:00 PM via cron job to the Synology NAS (Copy 1)
- **Weekly full backups** every Sunday to an external 4TB drive (Copy 2)
- **Monthly offsite rotation** of the second external drive to an employee's home or a safety deposit box (Copy 3, offsite)
- **Cloud backup** of critical source code and financial data to Google Drive or AWS S3 (additional offsite protection)

Reason: Startups rarely survive major data loss events. A single ransomware attack, hard drive failure, or office fire could destroy months of work if only one copy of data exists. Daily incrementals minimize the maximum data loss to one day of work. Weekly full backups provide a clean restore point that does not depend on a chain of incremental files. The offsite rotation protects against physical disasters like fire, flood, or theft. Cloud backup adds an extra layer of redundancy that protects against even catastrophic scenarios where both the NAS and external drives are lost simultaneously.

Security Recommendations

Recommendation: Implement a defense-in-depth security strategy with the following layers:

1. **Network-level security:** Deploy a pfSense firewall between the router and internal network to filter all inbound and outbound traffic. Configure firewall rules to block unnecessary ports and protocols.
2. **Network segmentation:** Create VLANs for each department (IT, HR, Finance, Sales) so that a security breach in one department cannot spread to others. Only the IT department's VLAN should have direct access to the server room network segment.
3. **Endpoint security:** Enable Microsoft Defender on all workstations, enforce BitLocker full-disk encryption via Group Policy, and ensure automatic Windows updates are enabled.
4. **Access control:** Restrict server access to the IT department using firewall rules and require VPN-only administration for remote management. Apply the principle of least privilege, meaning employees only get access to the systems they need for their job.
5. **Patch management:** Apply security patches monthly during a scheduled maintenance window to close known vulnerabilities before attackers can exploit them.

Reason: ABC Startup Solutions handles sensitive client source code, employee personal information, and financial data. A single data breach could destroy the company's reputation and result in legal liability. Segmentation contains a breach to a single department rather than exposing the entire network. Patching closes known vulnerabilities. Restricting administrative access reduces the attack surface and minimizes the risk of insider threats. Defense-in-depth means that even if one layer fails, the others continue to protect the organization.

Antivirus

Recommendation: Use Microsoft Defender as the baseline antivirus for all Windows 11 workstations, combined with Microsoft Defender Firewall (host-based) and the pfSense firewall (network-based). When the budget allows, upgrade to Microsoft Defender for Business, which adds Endpoint Detection and Response (EDR) capabilities for advanced threat hunting and incident response.

Reason: Microsoft Defender provides real-time, cloud-delivered protection at zero additional cost because it is included with Windows 11 Pro. Its behavioral monitoring detects suspicious patterns (like ransomware encrypting files) even when the specific malware strain is not yet in the signature database. Layering host-based protection (Defender on each workstation) with network-based filtering (pfSense) provides defense-in-depth. If malware slips past one layer, the other catches it. EDR capabilities become important as the company grows and faces more sophisticated threats that basic antivirus cannot detect.

Password Policy

Recommendation: Enforce the following password policy across the organization:

- **Minimum length:** 12 characters
- **Complexity:** Must include uppercase letters, lowercase letters, numbers, and special characters
- **Rotation:** Change passwords every 90 days, or immediately if a breach is suspected
- **Multi-Factor Authentication (MFA):** Enable on all email accounts, cloud services, and administrative portals
- **Password manager:** Require the use of Bitwarden (free, open-source) for storing unique passwords per system
- **Account lockout:** Lock accounts after 5 consecutive failed login attempts

Reason: Weak and reused passwords are the leading cause of account compromise worldwide. A 12-character password with mixed character types is virtually impossible to brute-force with current technology, even if the attacker has access to powerful hardware. MFA provides a second layer of protection that blocks credential theft even if a password is leaked in a data breach. A password manager encourages employees to use unique, complex passwords for every system without needing to memorize them, eliminating the dangerous habit of reusing the same password across multiple accounts. Account lockout prevents automated password-guessing attacks from running indefinitely against a user's account.

Expansion Plan

Recommendation: Design the network with room to grow from day one:

1. **Spare capacity:** The two 48-port switches provide 96 total ports for only 20 workstations, leaving 76 spare ports for future employees without purchasing new hardware.
2. **VLAN-ready infrastructure:** Managed switches and VLAN segmentation make it easy to add new departments or subnets without rewiring.

3. **Cloud migration path:** When user load exceeds the on-premises server's capacity, migrate applications to AWS or Azure. The Cloud Administrator should begin testing cloud deployment now so the transition is smooth when needed.
4. **Second-floor preparation:** Run extra CAT6 cable runs to the floor above during initial cabling installation, so expanding to a second office floor does not require ripping open walls later.
5. **Quarterly review:** Schedule infrastructure capacity reviews every 6 months to assess whether the current setup can handle growth, or if upgrades are needed.

Reason: A startup may double in size within a year. Purchasing a managed switch with extra ports and designing VLANs now costs very little compared to a full network redesign later. Virtualization and a defined cloud migration path allow the IT team to scale compute and storage without buying new physical hardware immediately. The cost of planning ahead is always less than the cost of scrambling to fix an under-provisioned network when the company is already growing.

PART 8 - Personal Reflection

When I first looked at this activity brief, I thought it would be straightforward: just pick computers, choose software, and draw a network diagram. I was wrong. The activity required understanding how every piece of technology connects to the business itself, and making sure every decision is justified not just technically, but financially and strategically.

The first thing I learned is that infrastructure planning is fundamentally about tradeoffs. As a startup, ABC Startup Solutions cannot afford enterprise-grade everything. I had to balance performance against cost, reliability against complexity, and current needs against future growth. For example, I chose a single tower server instead of a full rack setup because a 20-person company does not need redundant servers yet, but I made sure that server had ECC RAM and RAID 1 drives because losing the company's data would be catastrophic. That decision forced me to think like a businessperson, not just a technician.

The most challenging part for me was designing the network topology. Unlike the inventories, which mainly required research and logical thinking, the diagram forced me to think about how data actually flows through a network: how the internet enters the building, what path a file takes when an HR employee saves it to the shared drive, and how the firewall protects the server while still allowing employees to access the internet. It took several revisions before I felt confident the topology was technically accurate and easy for a manager to understand.

Another insight was how much I learned about the different IT roles by researching them. Before this activity, I thought "System Administrator" was a single job title that covered everything. Now I understand that a Helpdesk Technician, a Network Administrator, a Linux System Administrator, and a Cloud Administrator each have distinct skills, yet they must work together seamlessly: the Helpdesk Technician's ticket data reveals patterns that the Network Administrator uses to identify bottlenecks. Without all four, an IT team cannot function effectively.

This activity also made me deeply appreciate why planning must come before deployment. Without a hardware inventory, a startup might buy too many computers for one department and too few for another. Without a backup strategy, a single hard drive failure could destroy months of work. Without VLANs and firewall rules, a phishing attack could spread to the entire network. A good plan costs almost nothing on paper, but fixing a poorly planned network after the fact, in downtime, lost data, and emergency purchases, can be devastating for a small company.

Finally, I believe this activity has given me a foundation that will serve me throughout my career. I now approach technology from the perspective of business requirements rather than just technical specifications: the goal is not to build the most impressive network, but the right one for the company's mission, budget, and growth. I am proud that this work is now part of my portfolio, and I look forward to building on it in the weeks ahead.

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